Quantification: I534_z25c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	26
interior lacuna count	18
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 18)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2654.83	2139.5	1703.86	px^2
major_axis_length_px	lacuna shape	80.1708	74.6773	33.6413	px
minor_axis_length_px	lacuna shape	43.924	44.3137	14.2662	px
aspect_ratio	lacuna shape	1.8423	1.7303	0.5774	unitless
eccentricity	lacuna shape	0.7682	0.8145	0.1758	unitless
solidity	lacuna shape	0.8094	0.8235	0.1076	unitless
roots_count	headline	7.7778	6	4.4531	count
ring_length_r30_px	headline	253.167	253	110.357	px
ring_length_r60_px	reported	533.722	455	191.974	px
owned_length_px	ownership-dependent	541.71	403.363	402.36	px
edge_count	ownership-dependent	26.5556	21	19.7044	count
mean_edge_length_px	ownership-dependent	20.9968	20.343	5.9099	px
perimeter_px	normalised	262.234	255.978	100.818	px
ring_area_r30_px2	normalised	9200.56	9452	2733.38	px^2
ring_area_r60_px2	normalised	20062.3	20439	5163.94	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.992	1	0.0229	unitless
ring_density_r30	normalised	0.027874	0.026534	0.0102262	px^-1
ring_density_r60	normalised	0.0267588	0.0258261	0.00704439	px^-1
roots_per_100px_perimeter	normalised	3.0105	2.7003	1.4573	per 100 px
ring_attached_length_r30_px	network v2	216.611	202	113.081	px
ring_attached_length_r60_px	network v2	371.833	325.5	208.157	px
ring_length_w_r30_px	network v2	287.633	284.187	126.758	px
ring_length_w_r60_px	network v2	607.553	518.724	218.393	px
sholl_crossings_r10	network v2	5.9444	6	2.6451	count
sholl_crossings_r20	network v2	6.7778	6	3.766	count
sholl_crossings_r30	network v2	7.7778	7	3.1726	count
loop_count	loops, report only	1.3333	1	2.0864	count
loop_edge_share	loops, report only	0.1242	0.0963	0.1443	share
ring_width_mean_r30_px	width	7.6631	7.6405	0.8412	px

Per field
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measure	value	units
analysed_area_px2	842626	px^2
total_skeleton_length_px	22401	px
canalicular_length_density_per_px	0.0265848	px^-1
junction_count	439	count
junction_density_per_px2	0.00052099	px^-2
lacuna_count	26	count

lacunae_per_px2	3.086e-05	px^-2
skeleton_component_count	285	count
mean_component_length_px	78.6	px
median_component_length_px	27	px
field_length_density_w_per_px	0.0302208	px^-1
field_density_without_flagged_per_px	0.023029	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	225	px
unattached_component_count	16	count
canalicular_length_density_attached_per_px	0.0265848	px^-1
field_length_density_with_unattached_per_px	0.0268518	px^-1
excluded_area_px2	127816	px^2
analysed_field_area_px2	920760	px^2
loop_count	38	count
loop_edge_share	0.1016	share
edge_count	1221	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.8952	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	22269	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6611	188.219	59.1114	3.1841	0.9494	0.7758	-1.5406	20.9109	224.092	True
2	1077	56.963	28.6202	1.9903	0.8646	0.7016	0.9022	20.0604	411.138	True
3	10729	252.088	59.2099	4.2575	0.972	0.8231	-1.467	29.4163	690.039	True
4	7272	192.335	53.4975	3.5952	0.9605	0.8264	1.5113	27.3416	895.513	True
5	1175	57.981	35.1354	1.6502	0.7955	0.6155	-1.5521	82.0417	147.133	False
6	718	47.2395	23.002	2.0537	0.8734	0.8022	-1.0816	93.7911	437.035	False
7	2720	92.088	38.5852	2.3866	0.908	0.9547	-1.1428	215.941	986.257	True
8	506	33.0898	22.5781	1.4656	0.731	0.8447	-0.2309	223.868	412.781	False
9	6078	103.767	76.7807	1.3515	0.6727	0.9519	-0.9596	258.575	697.738	False
10	5138	135.156	50.3775	2.6829	0.9279	0.8874	-0.8946	375.376	523.812	False
11	749	51.0208	20.659	2.4697	0.9144	0.8874	1.1423	366.493	18.0734	True
12	2951	92.7411	57.1766	1.622	0.7873	0.6519	0.6626	412.099	797.415	False
13	3683	112.658	46.4293	2.4264	0.9111	0.7914	-0.9592	433.344	623.77	False
14	623	33.6165	26.9149	1.249	0.5991	0.8534	-1.4105	448.144	1010.89	True
15	3483	74.1218	60.6446	1.2222	0.575	0.9286	-0.204	483.984	66.1174	False
16	3784	85.9898	57.0703	1.5067	0.748	0.9309	-0.1766	493.888	417.965	False
17	4829	127.96	50.7336	2.5222	0.918	0.7542	-1.2747	583.146	228.783	False
18	2220	83.1379	43.4719	1.9125	0.8524	0.7754	0.0297	577.841	853.017	False
19	4318	144.296	45.1555	3.1955	0.9498	0.6869	1.4075	580.847	622.689	False
20	669	36.2679	35.241	1.0291	0.2363	0.673	1.4264	577.549	741.005	False
21	2037	75.2328	35.6451	2.1106	0.8806	0.9122	-0.2022	721.922	650.957	False
22	1103	54.1002	27.7589	1.9489	0.8583	0.9026	-0.7329	709.348	70.3019	False

23	812	45.0748	28.6241	1.5747	0.7725	0.7785	0.6665	738.431	13.2463	True
24	1992	69.0062	38.1155	1.8104	0.8336	0.9009	-1.2002	817.186	92.8996	False
25	1044	45.928	31.6235	1.4523	0.7252	0.8509	0.2821	817.663	289.875	False
26	2059	64.4016	53.6924	1.1995	0.5522	0.708	-1.4007	899.013	923.481	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	284	426	367.22	24	15.3008	7.4207
2	10	250	375	1596.95	79	20.2146	7.3007
3	8	259	583	157.196	10	15.7196	7.2057
4	3	184	418	134.125	6	22.3541	8.1476
5	9	234	406	209.782	17	12.3401	5.2679
6	2	193	414	68.4264	4	17.1066	8.307
7	3	55	210	61.7696	5	12.3539	6.4488
8	6	303	696	669.955	34	19.7046	7.8869
9	6	278	662	261.309	13	20.1007	7.5827
10	13	318	660	1008.68	49	20.5853	7.8377
11	3	126	307	183.61	6	30.6017	6.8573
12	13	460	1038	1583.83	76	20.8399	7.061
13	11	334	463	330.593	21	15.7425	7.3598
14	4	130	342	352.049	14	25.1463	7.5961
15	5	172	447	347.309	19	18.2794	8.499
16	4	151	431	180.095	10	18.0095	8.0738
17	18	496	814	820.411	50	16.4082	7.4463
18	7	272	601	771.625	32	24.1133	7.6143
19	13	311	542	467.747	21	22.2737	7.8907
20	11	204	369	386.25	23	16.7935	6.8089
21	3	147	421	420.475	11	38.225	8.8704
22	4	93	231	169.61	8	21.2013	7.4866
23	4	68	135	49.5563	5	9.9113	5.5824
24	6	190	379	585.73	24	24.4054	7.2164
25	3	106	415	278.321	9	30.9245	7.6667
26	6	295	618	1190.63	57	20.8883	9.0593

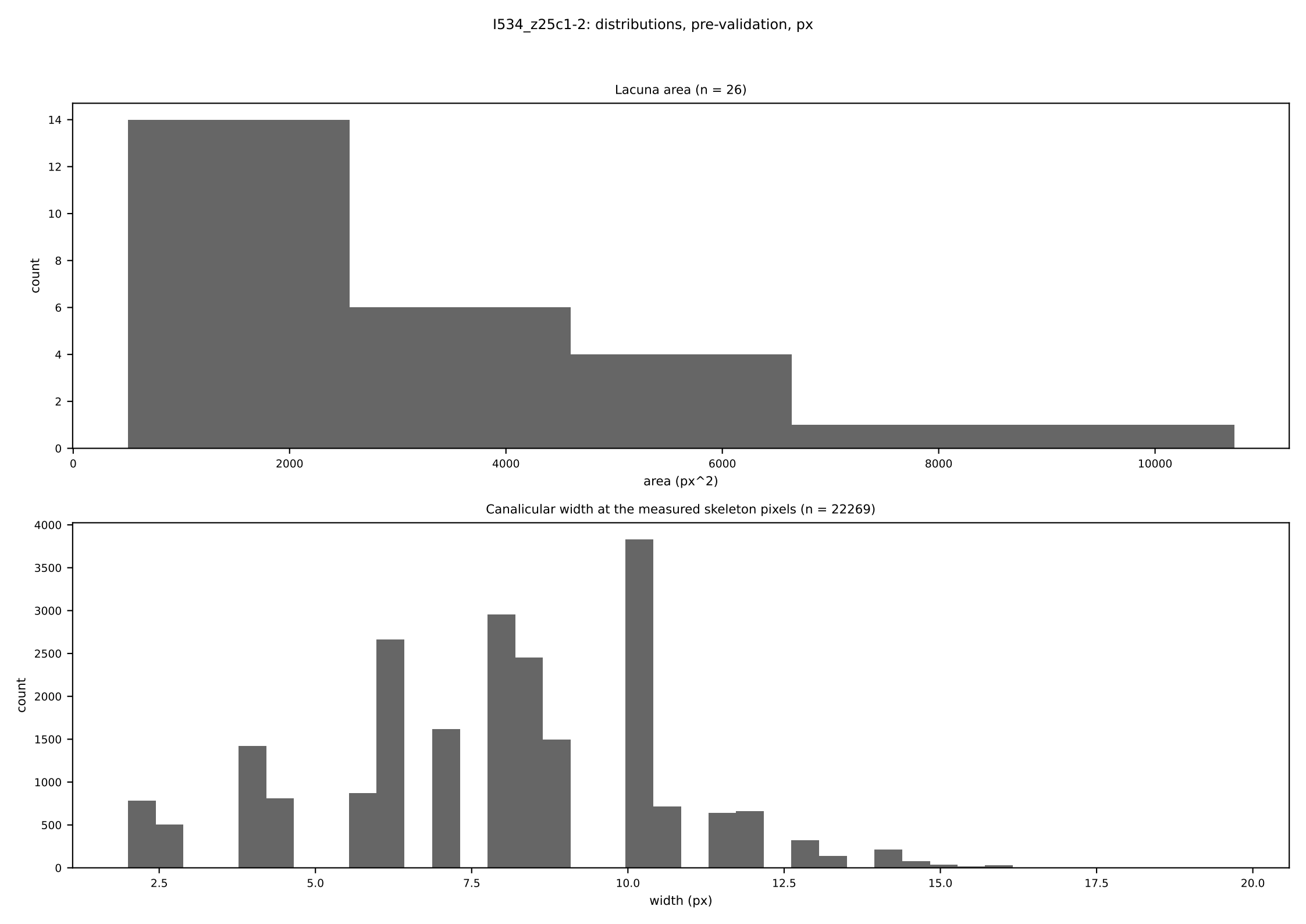
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	685.506	8694	14795	0.596	0.572	0.0326662	0.0287935	1.3129
2	192.167	4809	7186	0.6966	0.6322	0.0519859	0.0521848	5.2038
3	751.654	12022	23386	0.6838	0.6378	0.0215438	0.0249294	1.0643
4	629.754	9543	16652	0.6904	0.571	0.0192812	0.0251021	0.4764
5	206.965	6412	13348	1	0.9867	0.0364941	0.0304165	4.3486
6	132.125	5891	13464	1	1	0.0327618	0.0307487	1.5137
7	237.965	7174	17084	0.7631	0.6986	0.00766657	0.0122922	1.2607
8	107.154	5665	16255	1	1	0.0534863	0.0428176	5.5994
9	334.475	11891	25749	1	1	0.023379	0.0257097	1.7939
10	366.517	12817	22622	1	1	0.0248108	0.0291751	3.5469
11	128.947	4467	9758	0.694	0.6341	0.0282068	0.0314614	2.3265
12	419.345	11919	24709	1	1	0.0385938	0.042009	3.1001
13	332.676	11820	20074	1	1	0.0282572	0.0230647	3.3065
14	114.184	3796	10383	0.6248	0.5844	0.0342466	0.0329385	3.5031
15	249.078	9533	20117	1	0.9067	0.0180426	0.02222	2.0074
16	262.877	9914	22390	1	1	0.015231	0.0192497	1.5216
17	405.588	13364	31075	1	1	0.0371146	0.0261947	4.438
18	278.09	9508	20761	1	1	0.0286075	0.0289485	2.5172
19	406.416	12535	23662	1	1	0.0248105	0.0229059	3.1987

20	166.752	5502	11604	1	1	0.0370774	0.0317994	6.5966
21	220.35	8451	20862	1	1	0.0173944	0.0201802	1.3615
22	153.238	5825	12202	1	0.9659	0.0159657	0.0189313	2.6103
23	188.368	3275	4996	0.6244	0.5829	0.0207634	0.0270216	2.1235
24	215.037	8340	19166	1	0.9972	0.0227818	0.0197746	2.7902
25	146.853	6827	19239	1	1	0.0155266	0.0215708	2.0429
26	316.676	9396	23822	1	1	0.0313963	0.0259424	1.8947

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	253	303	324.179	489.103	6	7	6
2	230	355	282.38	430.262	4	6	6
3	232	387	297.007	664.328	5	7	5
4	125	176	213.995	482.416	5	5	8
5	203	220	265.137	460.99	9	4	4
6	179	354	215.924	466.818	7	5	5
7	55	55	66.1838	236.995	2	1	0
8	294	650	344.522	803.796	6	9	9
9	224	373	303.238	737.742	7	6	8
10	296	584	370.019	751.855	9	7	11
11	114	152	139.74	352.806	3	2	6
12	416	877	530.345	1180.76	10	16	16
13	324	453	392.161	534.487	6	9	9
14	109	303	145.397	390.191	2	3	2
15	125	276	191.196	502.961	3	7	6
16	64	125	167.054	486.233	2	4	6
17	480	694	566.801	924.879	11	15	9
18	211	392	308.179	701.21	6	9	11
19	248	365	344.522	609.73	7	6	10
20	201	254	227.267	415.12	4	2	6
21	97	179	169.196	489.073	4	6	6
22	76	125	111.225	266.764	2	4	5
23	52	52	70.0416	151.711	1	0	3
24	182	297	215.338	424.363	6	4	6
25	103	215	125.782	482.801	4	3	3
26	176	260	329.492	696.37	4	6	10


I534_z25c1-2: distributions, pre-validation, px
Lacuna area (n = 26)
count
area (px^2)
0
2
4
6
8
10
12
14
0
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 22269)
count
width (px)
0
500
1000
1500
2000
2500
3000
3500
4000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0